



Deep Learning in Asset Pricing

- Research question: can model changes improve asset-pricing performance?

Replication and Model Comparisons

Transformer, rolling windows, regimes and portfolio combinations

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Presentation overview

1. Main paper
2. Replication
- 3. Transformer**
- 4. Rolling windows**
- 5. Regime change**
- 6. Uncertainty**
7. Combination
8. Summary

- **Establish the published paper and replication benchmarks.**
- **Evaluate four extensions, then compare combined results.**

Main paper: model architecture and data

Goal: learn a stochastic discount factor through conditional pricing moments.

Component	Input and role	Output
Macro SDF encoder	178 macro series, LSTM	4 states
Pricing-weight network	46 characteristics + states	Two layers of 64 units
Macro instrument encoder	178 macro series, LSTM	32 states
Conditional network	Characteristics + states	8 bounded instruments

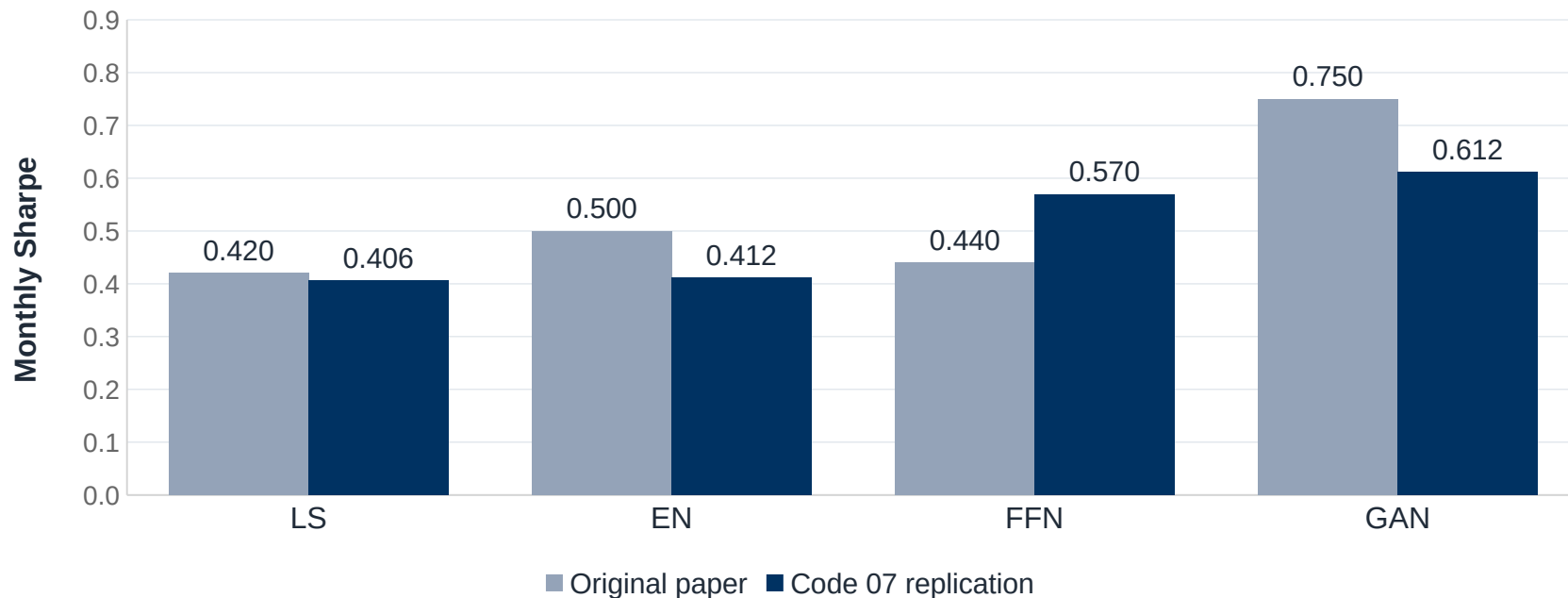
- The SDF network minimizes pricing errors, while an adversary exposes remaining errors.
- Train 1967–86, validate 1987–91, and test 1992–2016.

Main paper: published results

Model	Train	Validation	Test
LS	1.800	0.580	0.420
EN	1.370	1.150	0.500
FFN	0.450	0.420	0.440
GAN	2.680	1.430	0.750

- **GAN has the highest published monthly Sharpe in every split.**
- **Use these results as the reference benchmark for our replication and extensions.**

Replication: comparison with the main paper



- Our GAN test Sharpe is 0.612, compared with the paper's 0.750.
- Similar rankings support replication, but do not imply exact numerical reproduction.

Replication: validation and common signals

Check	Our result	Reference
FF-3 test Sharpe	0.196	Paper 0.190
FF-5 test Sharpe	0.223	Paper 0.220
GAN test correlation	0.800	Authors' monthly GAN
Leading characteristics	ST_REV, SUV, r12_2	Same top three

- **Factor Sharpe, return correlation and leading characteristics show economic similarity.**
- **Implementation differences can still affect the replication gap.**

Transformer: how the model changes

1. Represent the macroeconomic history

Project each month's 178 macro variables into a vector. Add position information so the model knows the order of months.

2. Attend to the available history

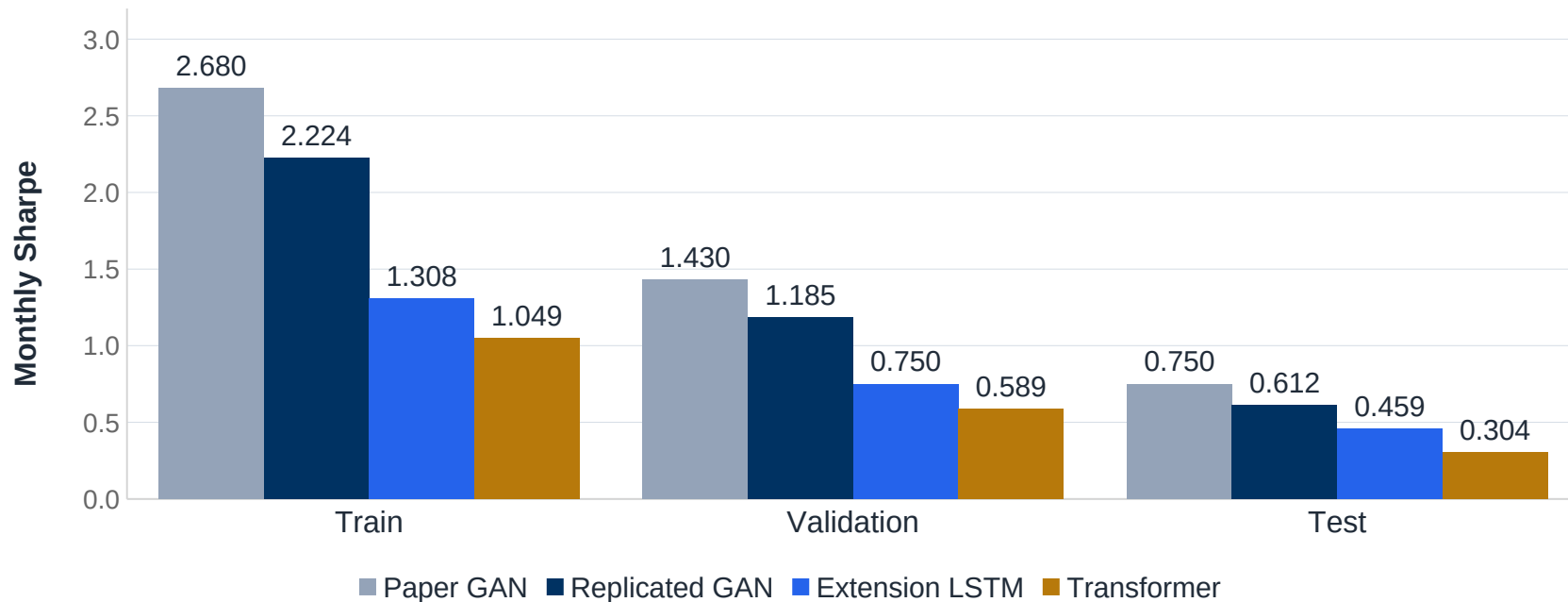
Self-attention learns which earlier months matter for the current state. A causal mask blocks access to future months.

3. Replace both LSTM encoders

Two separate Transformers output 4 pricing states and 32 adversarial states. The pricing and instrument networks retain their economic roles.

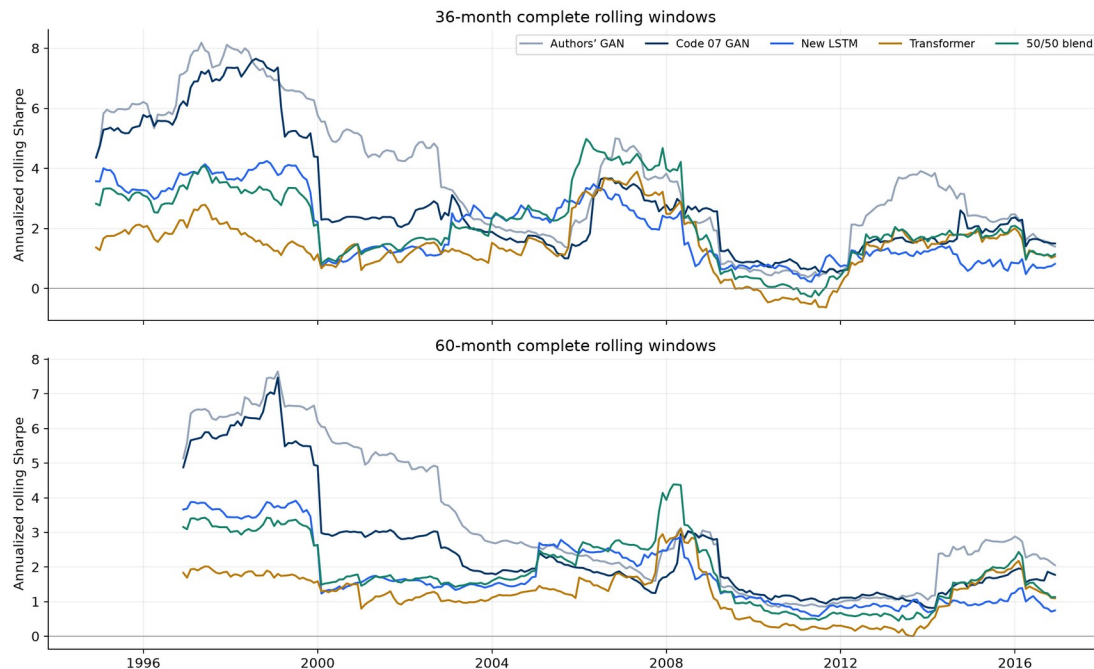
- **LSTM updates a recurrent memory. Attention directly weights available past months.**
- **The extension changes macro encoding while retaining the asset-pricing objective.**

Transformer: train, validation and test results



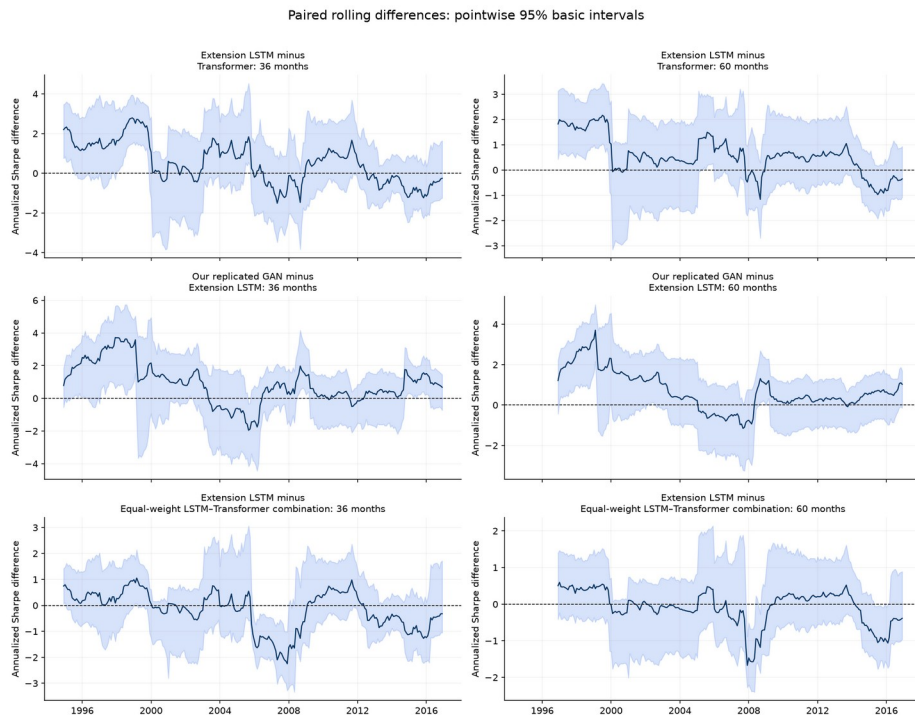
- **Extension LSTM leads Transformer: test monthly Sharpe is 0.459 versus 0.304.**
- **This comparison holds under our reduced budget, not the full author schedule.**

Rolling: 36- and 60-month Sharpe



- Curves show annualized Sharpe over the previous 36 or 60 months.
- Longer windows smooth fluctuations. These fixed models are not retrained.

Rolling: paired Sharpe differences



36- and 60-month windows

Line: observed annualized Sharpe gap.
Band: pointwise 95% basic interval.

- Line: observed paired Sharpe gap. Shading: pointwise 95% basic interval.
- A band crossing zero leaves that window's ranking unresolved.

Regimes: definitions and sample sizes

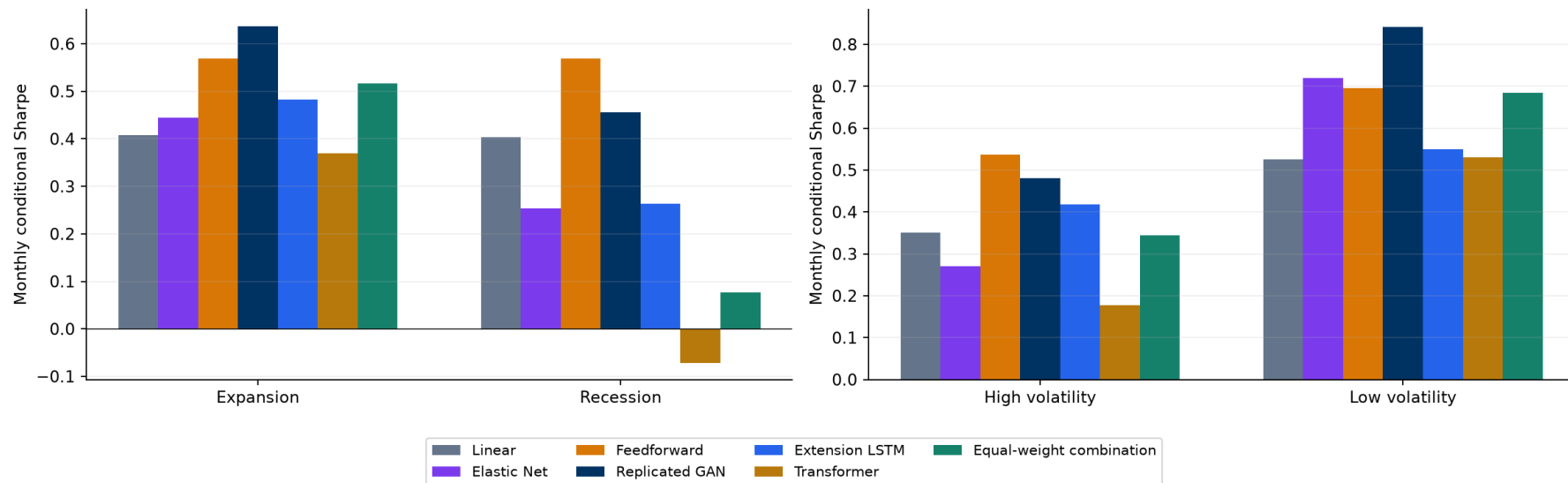
Regime family	Definition	Test months
Business cycle	NBER expansion / recession	274 / 26
Market volatility	High / low previous-12-month volatility	123 / 177

The volatility cutoff comes from the training sample. Recession dates are retrospective.

- **Business-cycle and volatility regimes classify different economic conditions.**
- **Only 26 recession months limit precision. NBER labels are retrospective.**

Regimes: comparison among our models

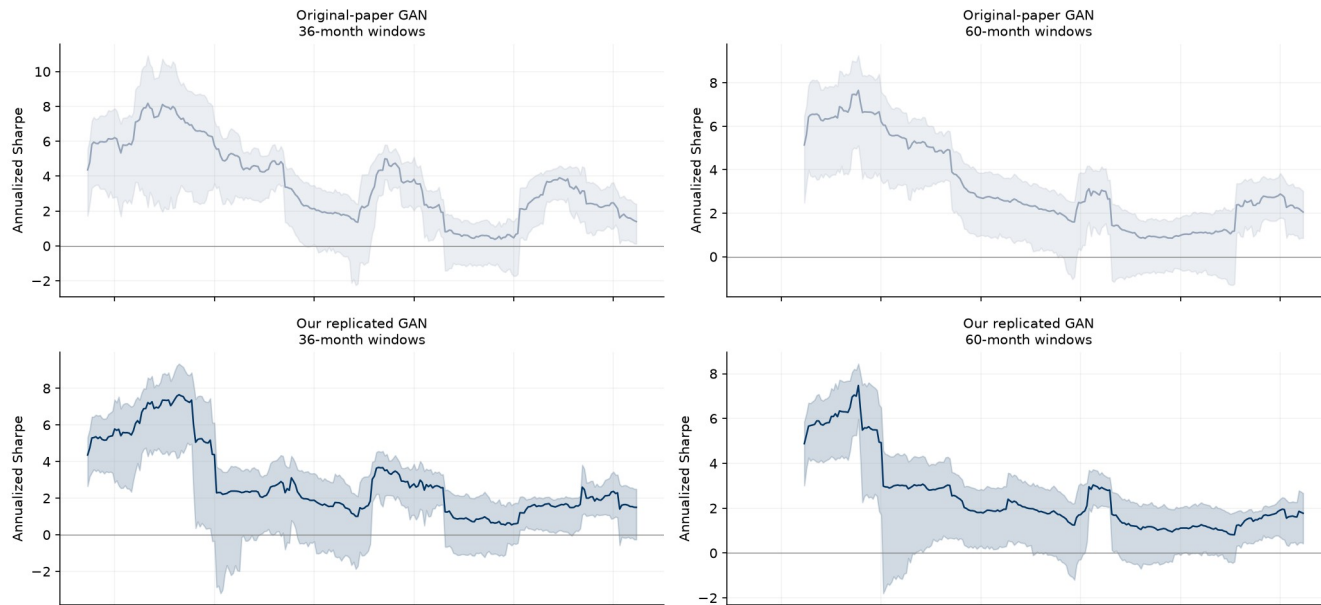
Local models only: business-cycle and volatility comparisons



- Bars show monthly Sharpe using only months in each regime.
- Feedforward leads recession Sharpe at 0.570, while Transformer has -0.071.

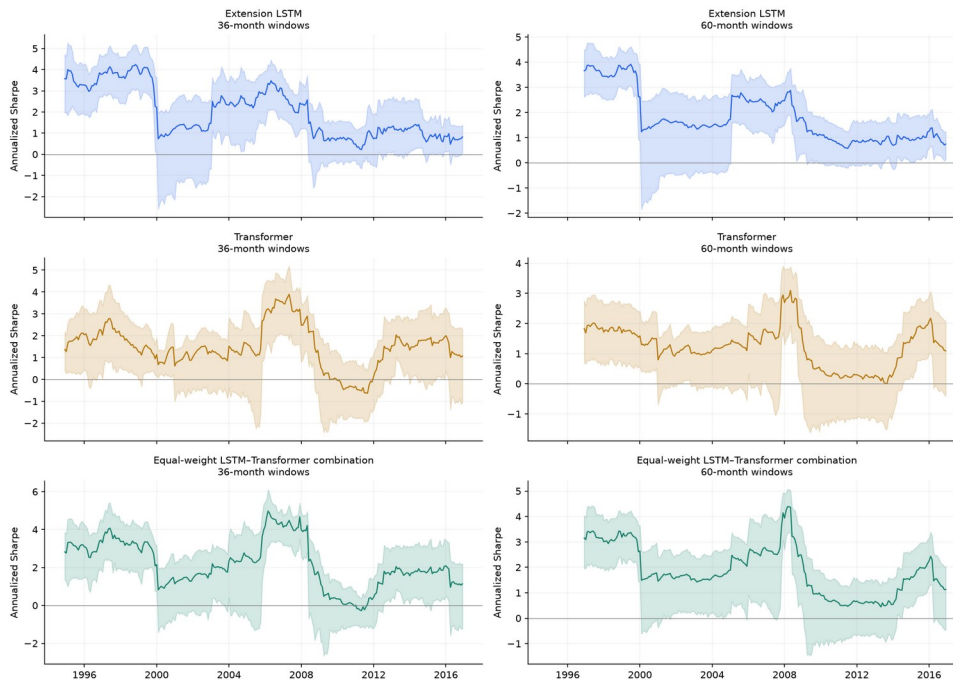
Uncertainty: paper and replication

Rolling model Sharpe: pointwise 95% basic intervals, six-month blocks



- Lines show annualized rolling Sharpe for the paper and replicated GAN.
- Some bands cross zero, leaving positive Sharpe unresolved in those windows.

Uncertainty: extension models



- Separate confidence bands show sampling uncertainty for each extension model.
- Several bands cross zero. Use the paired comparison on slide 15 to assess model gaps.

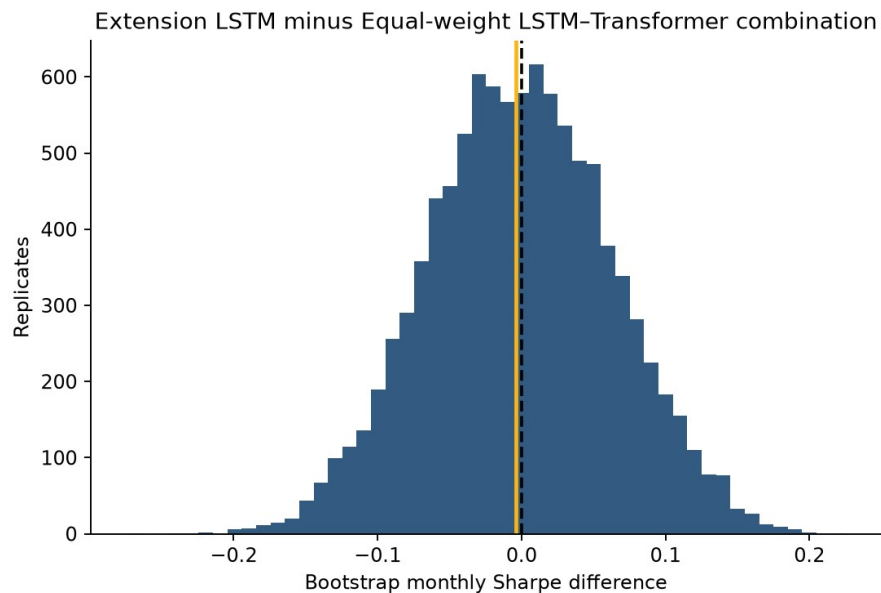
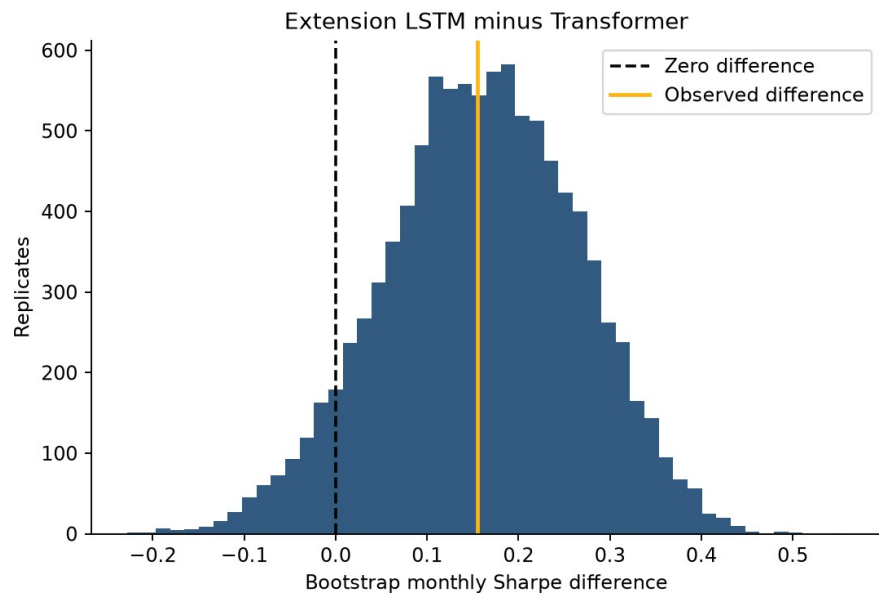
Uncertainty: paired Sharpe differences

A minus B	Difference	95% basic	95% simultaneous
Paper GAN minus Replicated GAN	0.138	[-0.037, 0.256]	[-0.148, 0.425]
Extension LSTM minus Transformer	0.155	[-0.049, 0.365]	[-0.132, 0.441]
Extension LSTM minus Equal-weight blend	-0.004	[-0.132, 0.119]	[-0.290, 0.283]

A positive point estimate can coexist with an interval containing zero.

- **LSTM exceeds Transformer by 0.155 monthly Sharpe, but both intervals include zero.**
- **The observed advantage remains statistically unresolved. This does not prove equality.**

Uncertainty: bootstrap Sharpe differences



- Bars count 10,000 draws. Gold marks the observed gap, black marks zero.
- Both paired 95% basic intervals include zero, despite different point estimates.

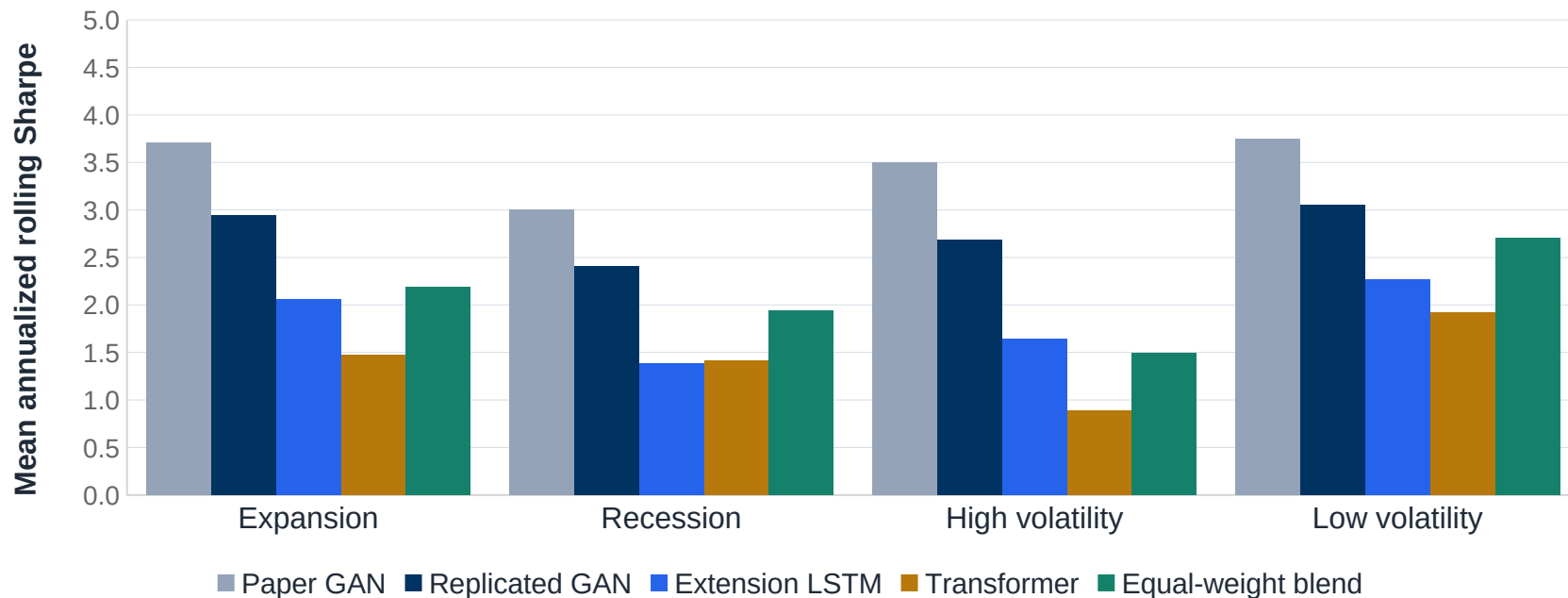
Combination: equal-weight architecture portfolio

Portfolio	Train	Validation	Test
LSTM	1.308	0.750	0.459
Transformer	1.049	0.589	0.304
50/50 blend	1.414	0.915	0.463

Test correlation between architectures is 0.292. Averaging lowers volatility.

- **Averaging saved LSTM and Transformer returns lowers volatility.**
- **Test Sharpe rises only from 0.459 to 0.463. The combination is exploratory.**

Combination: 36-month rolling results by regime



- Bars group rolling windows by the economic regime at their endpoint.
- A recession-ending window includes expansion months, unlike conditional regime Sharpe.

Combination: overall, rolling and recession results

Local portfolio	Test Sharpe monthly	36-month mean annualized	Recession Sharpe monthly
Replicated GAN	0.612	2.890	0.455
Feedforward	0.570	2.620	0.570
Equal-weight blend	0.463	2.166	0.076
Extension LSTM	0.459	1.992	0.264
Elastic Net	0.412	2.074	0.254
Linear	0.406	1.740	0.404
Transformer	0.304	1.467	-0.071

- **Overall, rolling and recession columns answer different questions and use different units.**
- **Replication leads local test Sharpe at 0.612. Feedforward leads recessions at 0.570.**

Summary: findings and next steps

What we add

A common-budget encoder comparison, rolling and regime evaluation, a fixed combination, and paired uncertainty.

What the evidence supports

LSTM has higher observed test Sharpe. Bootstrap intervals leave its advantage over Transformer unresolved.

What remains open

Full author-schedule GPU training, a new holdout, transaction costs and uncertainty across training runs.

- **Our contribution combines architecture, rolling, regime and uncertainty evaluation.**
- **Higher point estimates still need uncertainty checks and a new holdout.**